DH^{2}A-KT: Dynamic Heterogeneous Hypergraph and Agentic Knowledge Tracing with Leakage-Controlled Graph Construction

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Abstract*—Graph-augmented knowledge tracing (KT) models student mastery by propagating information over auxiliary concept graphs, but two gaps limit current practice. First, most graph-KT architectures represent only pairwise knowledge-component relations and omit heterogeneous entities – teachers, hints, forum discussions, instructional videos – that plausibly shape a learner's trajectory. Second, when graph-augmented KT models are evaluated without auditing whether the auxiliary graph was itself built without leakage, reported accuracy gains are difficult to attribute to genuine structural signal versus held-out information reaching the model through the graph channel. We present DH^{2}A-KT, a two-tier architecture that addresses both gaps while remaining computationally tractable on a single consumer GPU. Tier 1 (DH^2-KT) extends a leakage-controlled, train-only concept graph construction and audit protocol – released with a companion paper, P0 – from pairwise prerequisite/similarity edges to heterogeneous hyperedges (session, concept-prerequisite, discussion-thread, teacher-intervention), and adds a propensity-score causal-effect layer to separate genuine intervention effects from spurious correlation. We extend P0's four-diagnostic leakage taxonomy with three hyperedge-specific leakage classes – group-membership, cross-modal, and intervention-timing leakage – that a pairwise audit cannot detect. Tier 2 (AgentDG-KT) is a deliberately-scoped pilot in which a small team of LLM agents (Diagnostician, Tutor/Hint, Critic) reads Tier 1's frozen output to produce natural-language explanations, gated by a mandatory Critic agent rather than benchmarked at the same quantitative scale as Tier 1. We reuse P0's already audited, leakage-controlled baseline results on XES3G5M, ASSISTments 2012, and Junyi Academy as the Tier 1 comparison column, avoiding redundant retraining of nine baseline knowledge-tracing models. [TODO: Replace this sentence with the actual Tier 1/Tier 2 headline result once M5/M6/M9 (see docs/execution-plan.md) are complete – do not pre-announce a number before it exists.]

***Index Terms*—Knowledge tracing, hypergraph neural networks, heterogeneous graphs, large language model agents, causal inference, data leakage, educational data mining.**

I. Introduction

Knowledge tracing (KT) estimates a learner's latent mastery of knowledge components (KCs) from interaction history, supporting adaptive practice, intelligent tutoring, and curriculum recommendation. The field has progressed from probabilistic and sequence models [1] to graph-augmented models that propagate information over auxiliary concept graphs [2], [3], and, most recently, toward hypergraph representations that capture higher-order relations among knowledge components [4]–[6] and dynamic, temporally-evolving graph structure [7].

This paper addresses these three gaps directly, building on – rather than replacing – [8]'s leakage-control protocol. Tier 1 (Sections IV-A–IV-D) extends that protocol's train-only, audited concept-prerequisite edges into heterogeneous hyperedges and adds the causal-effect layer the first and third gaps call for; Section IV-B extends its four-diagnostic leakage audit to the hyperedge setting the second gap identifies as unaudited. Tier 2 (Section V) is a deliberately-scoped pilot showing that an LLM-agent interpretability layer can read Tier 1's frozen, audited output without reopening the leakage question Tier 1 already closed. **[TODO: Once M5/M6/M9 (docs/execution-plan.md) produce results, replace this paragraph's forward-looking framing with a results-grounded framing and add the C6 headline-finding sentence below – do not claim a gain exists before it is measured.]**

A. Motivation

Three concrete gaps motivate this work:

- **Heterogeneous relations are absent from graph-KT.** Hypergraph-KT models [4]–[6] group multiple knowledge components into a single hyperedge, but the hyperedge always contains only KC nodes. Teacher intervention, hint content, peer discussion, and instructional video – entities with documented influence on learning outcomes – are not represented as graph nodes in any published graph-KT model we are aware of.
- **Graph provenance is rarely audited.** [8] shows that pooling a knowledge-component graph from the full interaction log before a learner-based train/test split lets held-out information reach a graph-augmented KT model even when the sequence optimizer itself trains only on the training fold. [8]'s protocol audits this channel for *pairwise* prerequisite/similarity edges; no published audit protocol currently covers *hyperedges* spanning heterogeneous entity types.
- **Causal claims about pedagogical intervention are rare.** Most graph-KT models report predictive accuracy only,

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without attempting to separate a hint or teacher action's causal effect on subsequent correctness from confounded correlation.

B. Contributions

- **C1.** A heterogeneous hypergraph construction procedure (Section IV-A) that groups [8]'s already leakage-audited, acyclic prerequisite edges into concept-prerequisite hyperedges, and defines (interface-complete; data-blocked, see Section VIII-B) session, discussion-thread, and teacher-intervention hyperedge types.
- **C2.** A hyperedge-level extension of [8]'s four-diagnostic leakage taxonomy, adding three classes – group-membership, cross-modal, and intervention-timing leakage – that a pairwise audit structurally cannot detect (Section IV-B).
- **C3.** A causal-effect layer (Section IV-D) estimating the treatment effect of a hint or teacher intervention via inverse-propensity weighting, as a design-target starting point before a more elaborate causal architecture is attempted.
- **C4.** Direct reuse of [8]'s published, leakage-controlled baseline results (BKT, DKT, AKT, simpleKT, GKT, GIKT, SKT, DyGKT, DGEKT on XES3G5M/ASSISTments 2012/Junyi Academy) as the Tier 1 comparison column, avoiding redundant baseline retraining under matched protocol and split seeds.
- **C5.** A deliberately-scoped Tier 2 pilot (AgentDG-KT, Section V) demonstrating that an LLM-agent interpretability layer reading Tier 1's frozen output – rather than re-inferring knowledge state from scratch – is tractable on a single consumer GPU, gated by a mandatory Critic agent.

[TODO: Once results exist, add a C6 sentence stating the headline empirical finding (or its absence/negative result) – P0's own Introduction models this pattern (see its C5) and it should be mirrored here honestly, including if Tier 1's hypergraph gain turns out to be small or backbone-specific.]

II. Related Work

A. Graph- and hypergraph-based knowledge tracing

Early graph-augmented KT models propagate information over a single, pairwise concept graph: [2] layer a graph neural network over a knowledge-component graph on top of a sequence model, and [3] let exercises and concepts recursively re-rank one another through a bipartite exercise-concept graph. [7] extends this line to a temporally-evolving graph that updates edges as new interactions arrive rather than freezing the graph after construction. A second line moves from pairwise edges to hyperedges spanning more than two knowledge components at once: [4] proposes a temporal hypergraph memory network, [5] adds a dual-gated mechanism controlling how much a hyperedge's aggregated signal influences a given prediction, and [6] replaces graph convolution with a hypergraph transformer. A third line adds heterogeneity in what a node *represents* rather than how many nodes an edge spans: [9] augments the concept graph with psychological-factor nodes, and [10] builds a heterogeneous graph directly over learner-question interactions. Across all three lines, every published hyperedge or heterogeneous node we are aware of is still drawn from a closed set of KC-, exercise-, or learner-adjacent entities; none represent a Teacher intervention, Hint content, Forum discussion, or instructional Video as a first-class member inside the same hyperedge as the knowledge components it plausibly affects – the gap C1 (Section IV-A) addresses.

B. LLM agents in education and graph-augmented LLM agents

[11] surveys the growing use of LLM agents across tutoring, feedback generation, and curriculum planning in education, and [12] instantiates a multi-agent LLM pipeline that simulates computerized adaptive testing end to end, with agents standing in for both the test administrator and the examinee. Outside education, [13] surveys how LLM agents can read from and write to an external graph as working memory, and [14] applies a graph-retrieval-augmented-generation pattern to recommend personalized learning paths over a dual knowledge-structure graph. AgentDG-KT (Section V) differs from this pattern in one specific way: none of the above agents are gated by a design in which the graph – and the quantitative model that reads it – is frozen before the agent acts, with the agent limited to producing a natural-language artifact (rationale, hint, session hyperedge) that a separate Critic agent must approve before it is treated as anything more than a proposal. We adopt this frozen-core-plus-pilot-agent split specifically because a full multi-agent LLM pipeline that re-infers knowledge state end to end over millions of interactions is not tractable on a single consumer GPU (Section V), not because we believe agentic re-inference is undesirable in principle.

C. Leakage-controlled evaluation in knowledge tracing

[8] is the companion paper this work builds on and reuses directly (Section VI-A); we summarize its protocol here so the rest of this paper is self-contained. [8] observes that a knowledge-component graph is often constructed by pooling co-occurrence or similarity statistics over the *entire* interaction log before a learner-based train/test split is applied, which lets held-out information reach a graph-augmented KT model through the graph channel even when the sequence model itself trains only on the training fold. Its audit protocol reports four scalar diagnostics computed on a train-only-constructed graph – an edge-composition-ratio flag ($\mathrm{ECR}^{\mathrm{flag}}$), an edge-composition overlap ratio ($\mathrm{ECR}^{\mathrm{overlap}}$), an edge-outcome correlation ($|\rho|$), and a train/test boundary-mismatch rate (TBMR) – alongside a DAG audit with cycle pruning, a cold-start stratification of knowledge components into four frequency strata, a DAG Disruption Rate (DDR) manipulation check under five structured augmentation operators, and a ground-truth cross-validation against an expert-curated prerequisite DAG on Junyi Academy. Across XES3G5M,

ASSISTments 2012, and Junyi Academy, [8] reports that train-only graph construction shifts baseline AUC by at most 0.003 relative to full-log construction – evidence that leakage control, in their setting, does not by itself manufacture the reported gains, and a result we rely on directly when interpreting any Tier 1 gain in this paper (Section VI-A). Section IV-B extends this four-diagnostic protocol from pairwise edges to heterogeneous hyperedges.

D. Causal inference in educational data mining

Causal analysis in education has mostly targeted two separate questions: estimating the effect of a known treatment, and discovering an unknown causal structure among skills. [15] addresses the first: given historical course-enrollment and grade data, they estimate the average treatment effect of taking one course before another using matching and regression – directly analogous in spirit to the propensity-score estimator we use for hint/teacher-intervention effects (Section IV-D), though on a coarser course-level treatment rather than a step-level pedagogical action. [16] addresses the second: they learn a latent causal ordering and dependency structure among knowledge-component skills end to end inside a modified DKT model, evaluated on the NeurIPS 2022 Challenge on Causal Insights for Learning Paths in Education. [17] sits between the two, using a causal-relations-constrained student model to make KT predictions more interpretable without a separate treatment-effect estimation step. Our causal-effect layer (Section IV-D) is closest in spirit to [15] – a treatment-effect estimator rather than a causal-discovery method – but applies it at the level of individual hint/teacher-intervention hyperedges rather than curriculum-level course sequencing, and explicitly reports the propensity-overlap diagnostic that [15] does not discuss, following the identification-assumption caveat in Section VIII-B.

III. PRELIMINARIES

Let $\mathcal{D} = \{(u, t, q, c, y)\}$ be a KT interaction log with learner u, timestamp t, item q, knowledge component c, and correctness label $y \in \{0, 1\}$, following [8]'s notation. A learner-based temporal split $\mathcal{F} = (\mathcal{F}_{\text{train}}, \mathcal{F}_{\text{val}}, \mathcal{F}_{\text{test}})$ assigns each learner to exactly one fold (ratios $0.7/0.1/0.2$, three folds with split seeds $42/43/44$, matching [8]'s protocol so results remain directly comparable, see Section VI-A).

We extend this notation with a *hyperedge* $h = (\kappa, \{(\tau_i, e_i)\}_{i=1}^{k}, f)$: a kind label $\kappa \in \{\text{concept-prerequisite}, \text{session}, \text{discussion-thread}, \text{teacher-intervention}\}$, a set of k typed member entities (τ_i, e_i) with entity type $\tau_i \in \{\text{student}, \text{exercise}, \text{concept}, \text{teacher}, \text{hint}, \text{forum}, \text{discussion}, \text{video}\}$, and a fold identifier f. A heterogeneous hypergraph $\mathcal{H}_f = (\mathcal{V}_f, \mathcal{E}_f^{\text{hyper}})$ for fold f is a set of such hyperedges over the typed node set $\mathcal{V}_f$.

A. Design goal

For each fold f, the goal is a hypergraph $\mathcal{H}_f$ constructed using only $\mathcal{F}_f^{\text{train}}$ evidence, audited for the leakage classes in Table I (Section IV-B), that a downstream Tier 1 encoder consumes to predict y for held-out interactions.

IV. TIER 1: DH^2-KT

A. Heterogeneous hypergraph construction

[TODO: Describe build_concept_prerequisite_hyperedges() precisely (Algorithm IV-A below is a first draft matching dh2a_kt/hyperedge/construction.py – keep this in sync with the actual code as it is implemented per docs/execution-plan.md M1-M2).]

Input: audited, acyclic prerequisite edges E_{pre}^{f} from [8]'s Algorithm 2; chain length bounds $[\ell_{\min}, \ell_{\max}]$
Output: concept-prerequisite hyperedge set $\mathcal{E}_f^{\text{cprereq}}$
Build directed graph G from E_{pre}^{f}
for each root r with in-degree 0 in G **do**
  for each simple path p from r with $\ell_{\min} \le |p| \le \ell_{\max}$ **do**
    emit hyperedge with members $\{(\text{concept}, c) : c \in p\}$
  end for
end for

Session, discussion-thread, and teacher-intervention hyperedges are specified with the same interface but are *data-blocked* on all three core benchmarks: no public KT benchmark studied by [8] (XES3G5M, ASSISTments 2012, Junyi Academy) ships Hint, Forum, or Teacher-action logs synchronized to the interaction stream. We resolve this per hyperedge kind rather than uniformly. Discussion-thread and teacher-intervention hyperedges have no synchronized public source at all (Forum and Teacher-action logs are absent from every benchmark we are aware of); we scope these two hyperedge kinds to the Tier 2 case study (Section V) rather than the Tier 1 quantitative core. Session hyperedges (Student-Exercise-Concept-Hint(-Video)) have a candidate public source, ES-KT-24 [18], which ships educational-game-playing video alongside interaction logs. **[TODO: State whether ES-KT-24 wiring (docs/execution-plan.md M2) succeeded once attempted – if it does not, Tier 1's quantitative core remains scoped to concept-prerequisite hyperedges only, which is still sufficient to train DH^2-KT end to end per docs/idea-D-plan.md section 5.]**

B. Hyperedge-level leakage audit

[8] audits five leakage classes at the level of pairwise edge provenance (structural, edge, target, temporal, cold-start neighbourhood) using four scalar diagnostics: ECR^{flag}, $\text{ECR}^{\text{overlap}}$, $|\rho|$, and TBMR. These diagnostics are reused unmodified (Section VI-A) on the pairwise projection of a hyperedge set. Table I adds three classes that a pairwise audit cannot express because they require a *group* of heterogeneous members, not a single edge.

[TODO: Report actual leakage rates here once docs/execution-plan.md M1 runs on real XES3G5M fold data – Table I currently documents the diagnostic definitions only, not measured values.]

C. Encoding and hypergraph attention

[TODO: Section pending docs/execution-plan.md M3 (dh2a_kt/models/dh2_kt.py DH2KT.forward() implementation). Do not describe the attention mechanism in past

TABLE I
HYPEREDGE-LEVEL LEAKAGE TAXONOMY (EXTENDS [8] TABLE 1)

Class	Source	Diagnostic
Group-membership	A hyperedge with ≥ 1 member drawn from a held-out interaction while other members are train-only	compute_group_membership_leakage
Cross-modal	A Forum/Video embedding pooled over content timestamped after the prediction cutoff	compute_cross_modal_leakage
Intervention-timing	A Teacher/Hint hyperedge whose content was effectively chosen using knowledge of the student's later correctness	compute_intervention_timing_leakage

tense until it exists and has passed the sanity overfit test in M3's Definition of Done.]

D. Causal-effect layer

We estimate the average treatment effect (ATE) of a binary intervention (e.g., hint given) on next-step correctness via inverse-propensity weighting, following the identification assumption that intervention assignment is unconfounded given observed features – a strong assumption on observational KT data that we report as a limitation rather than a proven property (Section VIII-B). Given confounders X, treatment $T \in \{0, 1\}$, and outcome Y, the propensity score $\hat{e}(X) = P(T = 1 \mid X)$ is estimated by logistic regression, and

$$\widehat{\mathrm{ATE}} = \frac{\sum_i T_i Y_i / \hat{e}(X_i)}{\sum_i T_i / \hat{e}(X_i)} - \frac{\sum_i (1 - T_i) Y_i / (1 - \hat{e}(X_i))}{\sum_i (1 - T_i) / (1 - \hat{e}(X_i))}. \quad (1)$$

[TODO: Report propensity-overlap diagnostics alongside any ATE number, per the warning already implemented in dh2a_kt/models/causal_layer.py – an ATE without an overlap check is not publishable on its own.]

V. TIER 2: AGENTDG-KT PILOT

Tier 2 is deliberately scoped as a pilot, not a benchmark at Tier 1's quantitative rigor (docs/idea-D-plan.md, following the feasibility analysis in the accompanying project discussion: a full multi-agent LLM re-inference pipeline over millions of interactions is not tractable on a single consumer GPU within a realistic project timeline). Three agents operate on Tier 1's *frozen* output rather than re-inferring knowledge state:

- **Diagnostician Agent** produces a natural-language rationale for a given Tier-1 prediction, explicitly instructed not to override the given probability.
- **Tutor/Hint Agent** proposes the next hint/exercise and writes a new session hyperedge back into the graph (the agentic closed loop).
- **Critic Agent** is a *mandatory* quality gate that flags any Diagnostician output contradicting or ignoring the Tier-1 number it was meant to explain, before anything is treated as ground truth in a case study.

[TODO: Report pilot results (sample size, hallucination/flag rate, human evaluation if collected) here once docs/execution-plan.md M8-M9 complete. Do not report Tier 2 metrics using Tier 1's AUC/ACC framing – the two tiers use different evaluation standards by design (Section VI-C).]

VI. EXPERIMENTAL SETUP

A. Datasets

Following [8]'s benchmark-role assignment, XES3G5M [19] (18,066 learners, 7,652 items, 865 KCs, 6,413,353 interactions) is the *primary* benchmark: low consecutive-KC-repeat rate, non-saturated headline AUC, and the clearest graph-backbone separation among the corpora [8] studied. ASSISTments 2012 is secondary (single-skill Q-matrix; leakage-ablation sanity). Junyi Academy provides the ground-truth cross-validation role via its expert-curated prerequisite annotation.

B. Baselines reused from P0

We reuse [8]'s published baseline results (`baselines/p0_reused/baseline_results.csv`, 69 rows, and `baseline_fold_results.csv`, 205 per-fold rows; both copied verbatim from [8]'s `results/tables/`, pinned to commit `5e3d6a01c32a649f1655ac8712f03359bddfe478`, vendored 2026-08-02) for BKT, DKT, AKT, simpleKT, GKT, GIKT, SKT, DyGKT, and DGEKT under train-only, leakage-controlled graph construction, rather than retraining these nine models on a single RTX 3090. Reuse is valid only if our preprocessing exactly matches [8]'s (same hashed `kc_id` space, same Q-matrix). `scripts/01_verify_p0_preprocessing_match.py` checks that each DH^{2}A-KT dataset config points at an unmodified, checksummed copy of [8]'s own config and that the vendored commit is recorded; for XES3G5M this currently confirms the config file matches (sha256 `4f9f35d9...d980b2`) against the vendored commit above. **[TODO: This check only verifies the config file is present and pinned – it does not yet verify that the locally preprocessed `data/` tensors match P0's byte-for-byte (docs/execution-plan.md M0 gate); report that stronger check here once M0's raw data is downloaded and preprocessed.]**

C. Manipulation check

Following [8]'s Section 4.7 reasoning, we do not interpret any ablation result as evidence of hypergraph benefit unless DH^2-KT first passes a manipulation check: AUC must move meaningfully under near-total ($p = 0.90$) destruction of the constructed hyperedge set. A backbone whose AUC does not move under total destruction is *graph-inert*, and its ablation numbers carry no interpretable meaning about graph benefit. **[TODO: Report pass/fail and ΔAUC here once docs/execution-plan.md M6 runs.]**

VII. Results

[TODO: This entire section is empty by design (docs/execution-plan.md M5-M9 not yet run at the time this skeleton was drafted). Populate with: (1) Tier-1 AUC/ACC/NLL vs. reused P0 baselines, with 95% CI; (2) ablation table (per-node-type, hypergraph-vs-pairwise, static-vs-dynamic, with/without causal layer); (3) manipulation-check result; (4) ground-truth cross-validation F1 on Junyi; (5) Tier-2 pilot qualitative table. Do not fabricate placeholder numbers – leave this section unwritten until real results exist, per the anti-pattern warned against in every prior planning document for this project.]

VIII. Discussion

A. Threats to validity

Four risks are known ahead of any experiment and are stated here so they can be checked against, not discovered after the fact. First, Tier 1's baseline comparison (Section VI-A) is only valid if our preprocessing exactly reproduces [8]'s hashed `kc_id` space and Q-matrix; `scripts/01_verify_p0_preprocessing_match.py` checks the config level of this today (Section VI-A), but the data-tensor-level check remains open until docs/execution-plan.md milestone M0's gate is passed. Second, extending hyperedge construction beyond concept-prerequisite hyperedges may take materially longer than the session/discussion-thread/teacher-intervention interfaces suggest; if so, Section IV-A already scopes discussion-thread and teacher-intervention hyperedges to the Tier 2 case study rather than letting this risk block the Tier 1 quantitative core. Third, any epoch/batch-budget comparison against [8]'s published baselines that cannot be matched exactly will be reported as an *observational* rather than compute-matched comparison, following the same self-imposed limitation [8] applies to its own cross-corpus comparisons. Fourth, the manipulation check (Section VI-C) is a necessary, not sufficient, condition for interpreting an ablation result as genuine hypergraph benefit: a backbone that fails it is graph-inert and its ablation numbers are set aside, but passing it does not by itself establish that a specific hyperedge kind – rather than the hypergraph mechanism in general – is responsible for any observed gain. **[TODO: Once M5/M6/M7 results exist, replace this qualitative framing with the actual outcomes – did preprocessing verification pass at the data level, did any risk materialize, what was the compute-budget match status.]**

B. Limitations

- Session, discussion-thread, and teacher-intervention hyperedges remain data-blocked on all three core benchmarks; Tier 1's quantitative claims are scoped to Student-Exercise-Concept(-Hint-Video) unless Section IV-A's ES-KT-24 integration is completed.
- The causal-effect layer's identification assumption (no unmeasured confounding given observed features) is stated, not proven, on observational KT data.
- The hyperedge leakage audit's pairwise-projection approximation (Section IV-B) weights all projected pairs equally; a hyperedge-native statistic is future work.

IX. Conclusion and Future Work

[TODO: Write after Results/Discussion are complete.]

Acknowledgment

All Tier 1 experiments in this paper are designed to run on a single consumer-grade NVIDIA RTX 3090 (24GB), matching the compute budget already validated by [8] on the same benchmarks. **[TODO: Add funding source(s) and any human evaluators recruited for the Tier 2 pilot (Section V) once M8/M9 determine whether human evaluation was collected.]**

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